

Report on the manuscript entitled ‘Strain-coupled one-dimensional turbulence for rapid distortion’ authored by Sharma, Sparsh.

Under its present form, this very long (42 pages) manuscript is very difficult to read, with no clear take-home message(s) on its relevant and original results. Often, similar discussions and models took place in different sections (and plenty of sub- and subsubsections), with unuseful redundancy.

-) Section 2 illustrates what is the most difficult to understand in the core of this study, with almost disconnected sections 2.1 and 2.2. One-dimensional turbulence (ODT) is presented in subsection 2.1, with five subsubsections. This seems to be a follow-up of original studies by Kerstein, from really one-D molecular transport (a stochastic model?) to an attempt to mimic (3D) pressure redistribution and production by strain. Since the application to flows with mean shear has been published to JFM in 2001 by Kerstein *et al.*, I wonder if a clear review of the JFM paper should be better than the ms. subsection 2.1? The other subsection 2.2. begins with a very classical review of full Reynolds stress models with the linear and nonlinear pressure redistribution terms. The essentials for RDT follow for homogeneous anisotropic turbulence, very classically as well. The only apparent linkage between 2.1 and 2.2 is the reference to the ODT kernel (2-7)-(2-8). More generally, all details for solving RDT equations (2.14) by a tensorial kernel are gathered in the monograph by Sagaut & Cambon (2008 cited, a new edition 2018, Springer, is of particular interest with its more complete chapter 8 on turbulence with strain, namely.) This monograph will be referred to as SC2018, with many references therein, often more recent and more relevant than the ones of the ms..

-) What are really the objectives and the achievements of the study remain very difficult to size. In Eq. (3.1), two linear and local (?) operators seem to be some surrogates of ‘production’ and ‘pressure-strain’ operators. Further, a nonlinear operator is addressed, but without clear identification in the model. Near the end of the paper, before Eq. (5.4) a useful possible description of the model is given: *ODT evolves a velocity vector $\mathbf{u}'(y)$ on a single line aligned with the resolved direction, here taken along the contraction axis of the strain,...*

-) The main effort and the clearest part of the ms. is on the the RST pressure-strain operator and on its impact on RDT (exact for the linear part) and on the contribution of pressure to the RST equation (closure of linear $\Pi_{ij}^{(r)}$ and nonlinear $\Pi_{ij}^{(s)}$ tensors.) The related problem of closure appears in different sections, subsections, subsubsections, as follows.

* p. 8, 2.2. Oversimplified closures by Rotta and IP (Launder and coworkers) are mentioned.

* p. 11, 3.2.2. Exact form of $\Pi_{ij}^{(r)}$ from 3D spectral integration. Note that the nonlinear part can be found as well in SC2018, from the spectral integration of the spectral tensor of *third order* two-point velocity correlations.

* p.12, 3.2.3. RDT result for isotropic initial data, corresponding to an exact value (not tuned) of the constant in the IP closure.

* p. 16, 4.1.3. LRR ‘isotropization of the production’ model, Eq. (4.5)

* p. 35, 5.1. Addition of the slow part to the pressure equation, Eq. (5.1).

* p. 37, 5.3. Eqs. (5.7)-(5.10). Perhaps *the sole original result* for the closure of $\Pi_{ij}^{(r)}$?

-) Validation of (several versions of) the model is not clear. Why the author favours the old DNS by Lee and Reynolds (1985)? This is reported in a technical report, probably in connection with a CTR Summer program? I think that more recent achievements were carried out in further studies by Bill Reynolds and coworkers. None of the 15 figures of the ms. gives a convincing validation of a new model, capable of incorporating RDT and nonlinearity, vs. numerical or experimental results.

I have more incidental comments, as follows:

-) The introduction incorporates many references of studies of sound radiation, that are very far from the model developed in the ms.. Idem with the shock-turbulence interaction: as a single example, the effect of the shock significantly differs from a one-component RDT contraction of incoming turbulence (the theory of Ribner, 1954, is revisited with recent references and compared to RDT in SC2018.)

-) The last paragraph of the conclusions seems to open the way to an application to complex flow: ... *the distorted, anisotropic upwash spectrum at the leading-edge plane*, ... I think that such an objective merits to be supported, but this is not really done in the present ms..